

GlocalDoc: A Global and Local Information Bottleneck for Few-Shot Legal Document Representation

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Abstract—Modern language models produce contextual embeddings that capture local meaning with remarkable accuracy, yet they tend to lose track of a document considered as a whole. The limitation is acute for long and highly structured texts such as legal judgments, where the decisive signal is spread across many paragraphs. We present GlocalDoc, a pre-training framework that trains a paragraph-level encoder to produce document embeddings reflecting both local content and global structure. GlocalDoc couples a teacher-student architecture with an information bottleneck: a momentum teacher encodes the clean document, a student encodes a corrupted view, and a learned bottleneck compresses the pooled student representation while three alignment objectives match it to the teacher at the paragraph, partial-document, and full-document level. We evaluate GlocalDoc against a hierarchical masked language model baseline (H-MLM) and an unpretrained baseline on few-shot multi-label classification of European Court of Human Rights cases. Our working hypothesis was that a more global embedding should help most when supervision is scarce. The evidence supports this only in the extreme regime. With ten labelled examples per class GlocalDoc attains the best macro-F1 and, more tellingly, the lowest variance across seeds, while the masked language model often fails to learn at all. Once fifty or more examples per class are available the ordering reverses and H-MLM dominates. The bottleneck therefore behaves as a strong prior that is valuable under scarcity but discards lexical detail that becomes useful with more data. We also report a representational collapse specific to corpora with heavy boilerplate, and show that centring the teacher target removes it.

Index Terms—information bottleneck, self-distillation, document representation, few-shot learning, legal NLP, representational collapse.

I. Introduction

Pre-trained transformer encoders have made the local semantics of text easy to recover. A model such as RoBERTa [1] assigns nearly synonymous sentences nearly identical vectors, and this property underpins most retrieval and classification systems in production. The same models are far less reliable when the unit of interest is an entire long document. A single forward pass is bounded by a context window of a few hundred tokens, and the common workaround of encoding a truncated prefix throws away most of the text. Pooling sentence vectors recovers coverage but not structure, because mean

or maximum pooling treats a judgment as a bag of paragraphs and ignores how the parts compose into a decision.

Legal text makes the problem concrete. A European Court of Human Rights (ECtHR) judgment states facts across dozens of numbered paragraphs, and the articles that were allegedly violated depend on the interaction of those paragraphs rather than on any one of them. Annotating such documents is expensive and requires expert knowledge, so the realistic setting is few-shot: a handful of labelled cases per article, not thousands. A representation that already encodes the global shape of a document, before any task labels are seen, should transfer better to this regime than one tuned only for token-level prediction.

We study whether an explicit compression objective can produce such a representation. The information bottleneck principle [2] prescribes keeping the part of an input that is predictive of a target while discarding the rest, and its variational form [3] makes the idea trainable inside a neural network. We instantiate it for documents in a framework we call GlocalDoc. A student encoder reads a corrupted view of a judgment and pools it into a single vector through a stochastic bottleneck; a momentum teacher reads the clean judgment; and the student is trained to predict the teacher at three granularities while paying a compression cost on the bottleneck. The design borrows the asymmetry of non-contrastive self-distillation [4], [5], which is what prevents the encoder from satisfying the alignment objectives with a constant output.

We compare GlocalDoc to H-MLM, a hierarchical masked language model in the spirit of SMITH [6] that keeps the same encoder and the same document pooler but replaces compression and alignment with token reconstruction plus a single pooling target. Both are evaluated by freezing the pre-trained encoder, then fine-tuning a linear classifier on $N \in \{10, 50, 100\}$ labelled cases per article. The questions we set out to answer are the following.

- RQ1. Does global and local bottleneck pre-training yield better few-shot document features than hierar-

chical masked language modelling?

- RQ2. How does any advantage depend on the amount of supervision available at fine-tuning time?
- RQ3. What training instabilities does the objective introduce on a boilerplate-heavy corpus, and which mechanisms resolve them?

The contributions of the paper follow these questions. We describe the GlocalDoc objective and the stabilisation mechanisms it needs in practice. We report a controlled comparison in which the only variable is the pre-training objective, and we find a clear interaction with N : GlocalDoc wins at $N = 10$ and H-MLM wins from $N = 50$ onward. We show that the most useful property of GlocalDoc at $N = 10$ is not a large mean improvement but a sharp reduction in seed variance, where the baseline collapses to zero on several seeds. Finally we document a failure mode that is easy to overlook on natural images but severe on legal text: when documents are nearly identical in the encoder’s initial embedding space, the teacher target degenerates to a constant, and centring it in the style of DINO [7] restores a usable training signal.

II. Related Work

A. Information bottleneck and variational compression

The information bottleneck frames representation learning as a trade-off between keeping information about a target and compressing the input [2], [8]. The deep variational bottleneck makes this practical by parameterising a stochastic encoder and bounding the mutual information terms [3], which connects it directly to the variational autoencoder [9] and to β -VAE [10], where a weight on the Kullback-Leibler term controls the strength of compression. A recurring difficulty is posterior collapse, in which the encoder ignores the input and the bottleneck reverts to its prior. The free-bits heuristic [11] addresses this by exempting a small budget of information per latent dimension from the compression penalty. GlocalDoc uses a hard per-dimension form of free bits together with a warm-up on β , so that compression is introduced only after the alignment objectives have shaped the representation.

B. Non-contrastive self-distillation

Self-supervised methods that align two views of an input without negative pairs must avoid the trivial solution of a constant embedding. BYOL [4] and SimSiam [5] achieve this with an asymmetric design: a predictor network on one branch, a stop-gradient on the other, and in BYOL a slow moving-average (momentum) teacher in the lineage of MoCo [12]. DINO [7] adds centring and sharpening of the teacher output to keep it from concentrating on a single direction. VICReg [13] and Barlow Twins [14] take a complementary route and prevent collapse explicitly by regularising the variance and the cross-correlation of the embeddings. GlocalDoc combines these ideas rather than choosing among them, because on our corpus each one

alone proved insufficient, a point we return to in Section V and Appendix B.

C. Masked language models and long documents

BERT [15] and RoBERTa [1] established masked language modelling as the dominant pre-training objective for encoders, and DistilRoBERTa [16] provides a compact six-layer version that we adopt as the shared backbone. None of these handle long documents natively. Longformer [17] extends the context window with sparse attention, whereas hierarchical models such as SMITH [6] encode sentences or paragraphs independently and compose them with a second-level module. Our H-MLM baseline is hierarchical in this sense: it encodes each paragraph, trains an attention pooler to reconstruct the document from a partial view, and keeps token-level masked prediction as an auxiliary loss.

D. Legal NLP and few-shot classification

LexGLUE [18] collects several legal benchmarks, including the ECtHR allegation task derived from earlier work on judgment prediction [19]. The task is multi-label and strongly imbalanced, which makes macro-averaged F1 the appropriate primary metric. Few-shot text classification has been studied extensively for in-context learning with large decoders [20], but the encoder-plus-linear-head setting we use isolates the quality of the pre-trained representation from any prompting effect. Document and sentence embedding methods such as Sentence-BERT [21] and SimCSE [22] optimise pooled vectors directly, yet they target short inputs and do not model paragraph structure, which is the gap GlocalDoc addresses.

E. Balancing multiple objectives

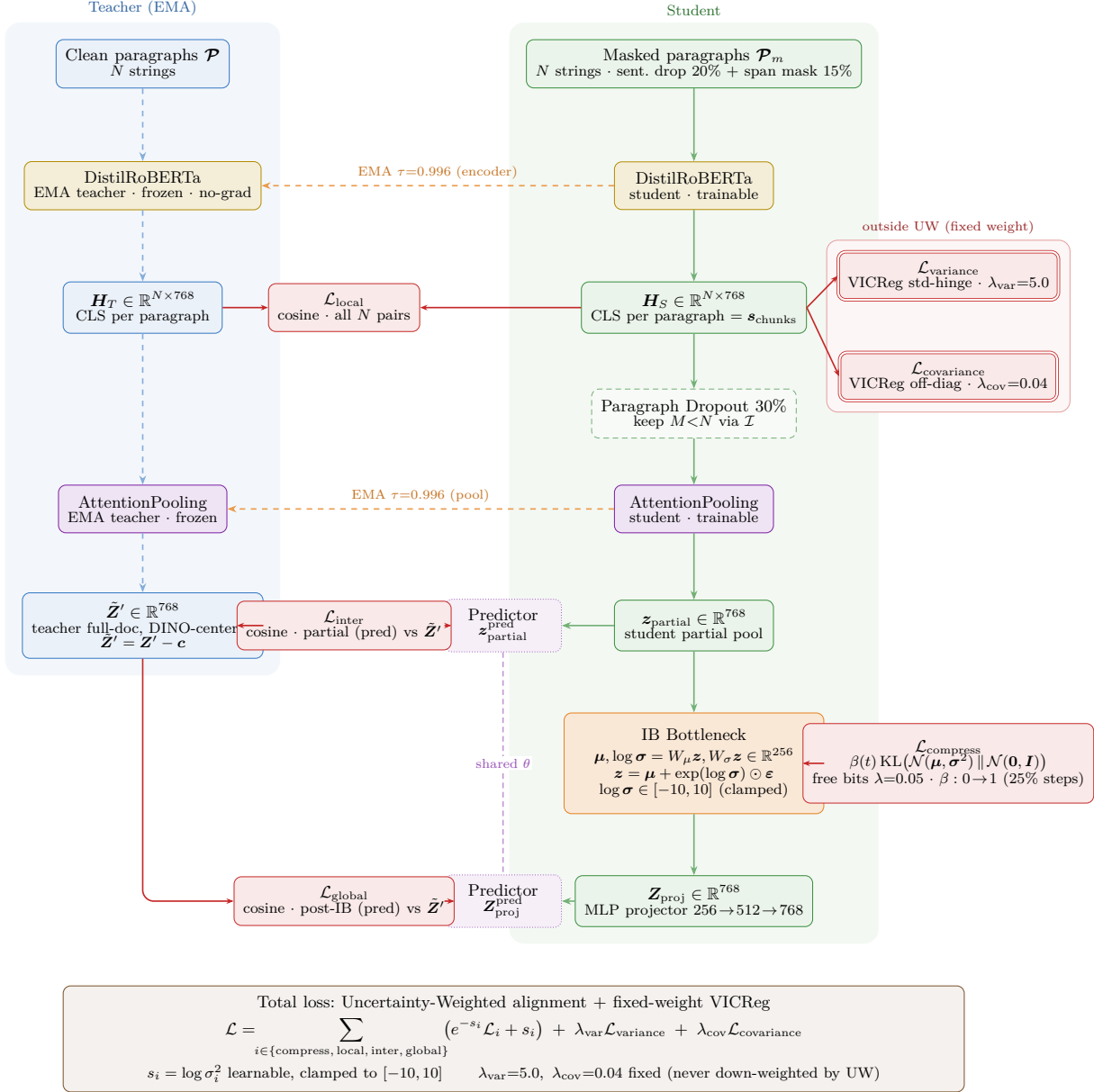
GlocalDoc optimises four losses at once. Rather than searching for fixed weights we use homoscedastic uncertainty weighting [23], which treats each loss as a Gaussian likelihood with a learned observation noise and produces self-tuning weights. We keep the anti-collapse regularisers outside this mechanism, since a signal whose purpose is to prevent degeneracy must not be down-weighted by the same optimiser that benefits from degeneracy.

III. The GlocalDoc Framework

Figure 1 summarises the architecture. We first fix notation, then describe the encoder, the two-level masking, the bottleneck, the alignment losses, and the mechanisms that keep training stable.

A. Notation and hierarchical encoding

A document is a sequence of paragraphs $x = (p_1, \dots, p_N)$ with $5 \leq N \leq 50$. Each paragraph is encoded independently by a shared transformer f_θ , taking the [CLS] representation as the paragraph vector. Paragraphs longer than 510 tokens are split into non-overlapping sub-chunks, each encoded separately, and their [CLS] vectors are averaged, so that a paragraph of any length maps to



— trainable (student) — EMA teacher (no-grad) — EMA update (encoder + pool), $\tau=0.996$ — loss signal
 $[\dots]$ Predictor: student-only BYOL/SimSiam head 768 → 512, LN, GELU, → 768 (shared θ ; teacher has none) $\boxed{}$ double border = fixed weight, outside UW
Training also applies temporal VICReg on $Z_{\text{proj}}^{\text{pred}}$ and μ via rolling buffers (script-level regulariser; not shown).

Fig. 1. The GlocalDoc architecture. The teacher branch (left, blue) is a frozen exponential moving average of the student: it encodes the clean paragraphs with DistilRoBERTa, pools them, and centres the full-document target as $\tilde{Z}' = Z' - c$ to remove the constant boilerplate direction. The student branch (right, green) encodes a masked view, drops 30% of paragraphs at the pooling stage, compresses the partial pool through the information bottleneck, and projects it back to $\mathbb{R}^{768}$. A single student-only BYOL/SimSiam predictor with shared weights θ is applied to both the partial pool and the projected code. The loss spine (centre) holds the three cosine alignment terms, $\mathcal{L}_{\text{local}}$ between paragraph matrices and $\mathcal{L}_{\text{inter}}, \mathcal{L}_{\text{global}}$ against the centred teacher target, together with the compression term $\mathcal{L}_{\text{compress}}$, all balanced by uncertainty weighting. The VICReg variance and covariance regularisers (right) act on the student paragraph vectors with fixed weights, outside the uncertainty-weighted sum. Temporal VICReg on $Z_{\text{proj}}^{\text{pred}}$ and μ is applied at the script level and omitted here.

one vector in $\mathbb{R}^{768}$. Stacking gives the paragraph matrix $S = [s_1; \dots; s_N] \in \mathbb{R}^{N \times 768}$.

A document vector is produced by an attention pooler

$$a_i = \frac{\exp(q^\top (s_i + e_i))}{\sum_j \exp(q^\top (s_j + e_j))}, \quad \text{pool}(S) = \sum_i a_i (s_i + e_i), \quad (1)$$

where q is a learned query and e_i are learned positional embeddings. The positional embeddings are initialised to zero, so the pooler begins as exact mean pooling and departs from it only as training requires.

B. Two-level masking

GlocalDoc corrupts the input at two independent levels. At the text level, each paragraph is masked before being read by the student: sentences are dropped with probability 0.20, and within the surviving text spans of three to five words are replaced by a single mask token until about 15% of words are covered. At the structural level, a paragraph dropout removes 30% of paragraphs, but only at the pooling stage. The student encoder still sees all N masked paragraphs; the dropout selects a subset of the resulting vectors before pooling, which yields a partial view $z_{\text{partial}} = \text{pool}(S_{\text{kept}})$. Keeping all N paragraphs in the encoder is required so that the paragraph-level objective can compare student and teacher vectors position by position.

C. Information bottleneck

The partial document vector is passed through a stochastic bottleneck. Two linear heads produce a mean and a log standard deviation, and a latent code is sampled with the reparameterisation trick,

$$\mu = W_\mu z_{\text{partial}}, \quad \log \sigma = \text{clip}(W_\sigma z_{\text{partial}}, -10, 10), \quad (2)$$

$$z = \mu + \sigma \odot \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, I), \quad (3)$$

with $z \in \mathbb{R}^{256}$. A two-layer projector g maps z back to $\mathbb{R}^{768}$, giving $Z_{\text{proj}} = g(z)$. The compression cost is the Kullback-Leibler divergence to a standard normal prior, computed in closed form and floored per dimension by a free-bits budget λ ,

$$\mathcal{L}_{\text{compress}} = \beta_t \sum_{j=1}^{256} \max\left(\lambda, \frac{1}{2}(\mu_j^2 + \sigma_j^2 - 1 - 2 \log \sigma_j)\right). \quad (4)$$

The clip on $\log \sigma$ keeps the term finite under bfloat16. The weight β_t follows a linear warm-up from zero to one over the first quarter of training, which lets the alignment objectives organise the representation before compression starts to bite. We use $\lambda = 0.05$ nats per dimension. A larger floor was tried and rejected: at $\lambda = 0.5$ the 128-nat floor dominated the objective and forced the mean toward a fixed magnitude regardless of the input, which is the opposite of the intended behaviour.

D. Multi-granularity alignment

The teacher is a momentum copy of the encoder and pooler, never updated by gradient. It reads the clean document and produces a full-document target $Z' = \text{pool}_T(S_T)$, which is centred as described below. Writing sg for stop-gradient, h for the predictor, and $d(u, v) = 1 - \cos(u, v)$ for the cosine distance, the three alignment losses are

$$\mathcal{L}_{\text{local}} = d(S, \text{sg}(S_T)), \quad (5)$$

$$\mathcal{L}_{\text{inter}} = d(h(z_{\text{partial}}), \text{sg}(Z')), \quad (6)$$

$$\mathcal{L}_{\text{global}} = d(h(Z_{\text{proj}}), \text{sg}(Z')). \quad (7)$$

The local term matches every student paragraph vector to the corresponding teacher paragraph vector. The intermediate term asks the partial pooled view to predict the full document before compression, and the global term asks the compressed and projected code to predict the same target after compression. Both document-level terms regress onto a single teacher vector, so the bottleneck must preserve whatever is needed to reconstruct the global representation from a partial and corrupted input.

E. Preventing collapse

The predictor h is applied to the student branch only. This asymmetry, together with the stop-gradient on the teacher, is the formal mechanism that removes the constant-output fixed point in non-contrastive self-distillation [4], [5]. The teacher tracks the student through an exponential moving average with rate $\tau = 0.996$, applied to both encoder and pooler after every optimiser step; running statistics in normalisation layers are copied rather than averaged, since averaging them drifts away from the student’s activation distribution.

On its own the asymmetry was not enough on legal text. We add the variance and covariance regularisers of VICReg [13] on the student paragraph vectors,

$$V(S) = \frac{1}{d} \sum_j \max(0, \gamma - \sqrt{\text{Var}(S_{\cdot j}) + \epsilon}), \quad (8)$$

$$C(S) = \frac{1}{d} \sum_{i \neq j} [\text{Cov}(S)]_{ij}^2, \quad (9)$$

with the variance hinge $\gamma = 0.5$. Because per-document paragraph statistics do not see across-document variation, we add the same two regularisers in a temporal form on the post-predictor document code and on the bottleneck mean, computed against a rolling buffer of the sixteen most recent samples whose gradient is detached. These temporal terms penalise a representation that is constant across consecutive documents even when it varies within a document.

The remaining and most important fix concerns the teacher target itself. ECtHR judgments share a large amount of boilerplate, and in the encoder’s initial space the documents are nearly indistinguishable: a probe over

200 documents found a median pairwise cosine of 0.9997, with every ordered pair above 0.99. A teacher target that is almost constant across documents lets the predictor satisfy $\mathcal{L}_{\text{inter}}$ and $\mathcal{L}_{\text{global}}$ by emitting that constant. We remove this fixed point with DINO-style centring [7]: a running mean c is maintained with momentum 0.9 and subtracted from the teacher output,

$$Z'_c = Z' - c, \quad c \leftarrow 0.9c + 0.1 \overline{Z'}_{\text{batch}}. \quad (10)$$

The centred vector is the actual regression target. Section V shows that this single change is what keeps training stable.

F. Objective and optimisation

The four primary losses are combined by homoscedastic uncertainty weighting [23] with learned log-variances $s = (s_c, s_l, s_i, s_g)$,

$$\mathcal{L}_{\text{UW}} = \sum_k (e^{-s_k} \mathcal{L}_k + s_k), \quad (11)$$

and the anti-collapse regularisers are added with fixed weights outside the weighted sum,

$$\mathcal{L} = \mathcal{L}_{\text{UW}} + \omega_v V(S) + \omega_c C(S) + \mathcal{L}_{\text{temporal}}. \quad (12)$$

The weights s are given their own parameter group with a learning rate three orders of magnitude larger than the encoder and no weight decay. Without this, the gradient on s is too small to move the weights away from their initialisation, and the uncertainty weighting stays degenerate at uniform weights for the whole run.

G. The H-MLM baseline

H-MLM keeps the backbone, the paragraph encoding, and the attention pooler, and removes everything specific to GlocalDoc. It optimises a token-level masked language modelling loss $\mathcal{L}_{\text{mlm}}$ on each paragraph with a 15% mask rate, and a single pooling objective that asks the attention pool over a 70% subset of paragraphs to predict the stop-gradient mean of all paragraph vectors,

$$\mathcal{L} = \frac{1}{2} \mathcal{L}_{\text{mlm}} + \frac{1}{2} d(\text{pool}(S_{\text{kept}}), \text{sg}(\overline{S})). \quad (13)$$

There is no bottleneck, no teacher, no predictor, and no uncertainty weighting. The comparison between GlocalDoc and H-MLM is therefore a comparison of pre-training objectives under an identical architecture and an identical fine-tuning procedure.

IV. Experimental Setup

A. Data

We use the ECtHR Task A subset of LexGLUE [18], in which the input is the list of fact paragraphs of a case and the target is the set of allegedly violated Convention articles over ten labels. We keep documents with between five and fifty paragraphs, which removes fewer than one percent of cases and bounds the cost of encoding. The label distribution is severely skewed. The most frequent

TABLE I
Few-shot test results on ECtHR Task A. Mean over five seeds, standard deviation in parentheses. Best per row block in bold; the best method at each N is the bold value within that N .

Method	N	Macro-F1	Micro-F1
No pre-training	10	0.001 (0.001)	0.003 (0.004)
	50	0.025 (0.008)	0.077 (0.030)
	100	0.042 (0.013)	0.106 (0.019)
H-MLM	10	0.019 (0.023)	0.069 (0.085)
	50	0.273 (0.061)	0.354 (0.060)
	100	0.372 (0.032)	0.454 (0.030)
GlocalDoc	10	0.044 (0.009)	0.139 (0.045)
	50	0.108 (0.024)	0.224 (0.027)
	100	0.183 (0.006)	0.306 (0.014)

article appears in roughly 4,700 training cases and the rarest in about 41, a ratio close to one hundred. This motivates macro-averaged F1 as the primary metric, with micro-averaged F1 reported alongside it.

B. Pre-training

Both objectives pre-train DistilRoBERTa [16] on the same filtered training split. GlocalDoc runs for five epochs and H-MLM for three, both with batch size one and gradient accumulation, in bfloat16 with a linear schedule and ten percent warm-up. The full configuration is listed in Appendix C, Table II. We deliberately report a single pre-training run per objective, which is a limitation we discuss in Section VI.

C. Fine-tuning protocol

At fine-tuning time the pre-trained encoder and pooler are loaded into a document classifier with a linear head over the pooled vector. For GlocalDoc we load the student pooler, since the teacher pooler is a moving average that barely departs from its initialisation over a short run. For each condition we evaluate $N \in \{10, 50, 100\}$ labelled cases per article, sampled with a multi-label aware routine that keeps the per-article quota and deduplicates cases covering several articles. Each setting is repeated over five seeds. The seed is fixed before both the few-shot sampling and the construction of the classifier head, so that a seed varies both the sample and the head initialisation rather than the sample alone. We train for ten epochs with a binary cross-entropy loss over the ten labels, a cosine schedule with ten percent warm-up, and a decision threshold of 0.5 at evaluation. The checkpoint with the best validation macro-F1 is evaluated on the held-out test split.

D. Baselines

Besides H-MLM, we include a no-pre-training baseline that loads an off-the-shelf DistilRoBERTa with a freshly initialised pooler. It measures how much of the few-shot performance is attributable to pre-training at all, as opposed to the architecture and the fine-tuning procedure.

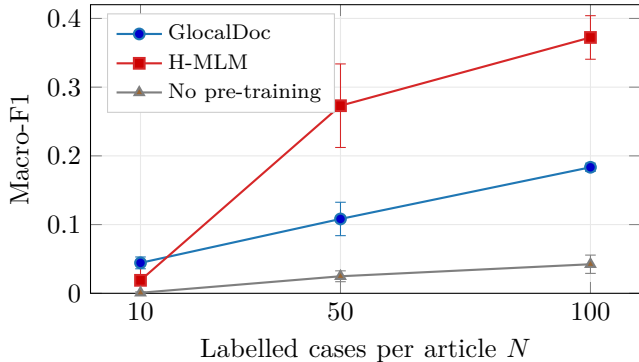


Fig. 2. Macro-F1 against the number of labelled cases per article, with one standard deviation over five seeds. GlocalDoc leads at $N = 10$ and has the smallest spread there; H-MLM overtakes it from $N = 50$ and widens the gap at $N = 100$.

V. Results and Analysis

Table I reports the test results and Figure 2 plots the macro-F1 against N . We read them against the three research questions.

A. RQ1 and RQ2: the advantage is regime-dependent

There is no single winner. At $N = 10$ GlocalDoc reaches a macro-F1 of 0.044 against 0.019 for H-MLM and 0.001 for the unpretrained baseline, and the same ordering holds for micro-F1. From $N = 50$ the picture inverts: H-MLM reaches 0.273 against 0.108 for GlocalDoc at $N = 50$, and 0.372 against 0.183 at $N = 100$. The hypothesis that a global bottleneck representation helps under scarcity is therefore confirmed only in the smallest regime, and is contradicted once a moderate amount of supervision is available.

The most informative quantity at $N = 10$ is the variance rather than the mean. The per-seed scores in Appendix D, Table III, show that H-MLM returns 0.000 and 0.001 on two of five seeds at $N = 10$, that is, it fails to learn anything on those runs, while GlocalDoc stays in a narrow band between 0.035 and 0.057. The bottleneck behaves as a strong prior. When ten cases per article are not enough to identify a good linear separator from a high-variance MLM representation, the compressed representation still supports a usable one, and it does so consistently. As N grows the prior turns into a handicap, because the same compression discards lexical detail that the MLM objective preserves and that a linear head can exploit once it has enough labelled examples to do so. The unpretrained baseline stays near zero throughout, which confirms that the effects we measure come from pre-training and not from the architecture or the optimiser.

B. RQ3: collapse and the role of centring

GlocalDoc does not train out of the box on this corpus. Before the centring fix, the inter-document cosine of the student document code climbed to about 0.99 within the first few hundred steps and the run produced the

degenerate representation already visible in the early baseline numbers. Figure 3 shows the first epoch after the fix. The collapse metric, defined as the mean pairwise cosine of the post-predictor document codes over a sliding window, falls from 0.42 to below 0.05 and stays well under the alarm and stop thresholds for the rest of the epoch (panel a). The centred teacher target has a pairwise cosine near zero throughout, which is direct evidence that centring absorbs the constant component of the boilerplate-dominated representation. The bottleneck stays informative: the fraction of latent dimensions above the free-bits floor rises toward 0.58 as β warms up, and the mean magnitude of μ grows in step (panel b). The two document-level alignment losses decrease over the epoch despite the rising compression weight (panel c). The uncertainty weighting is active rather than degenerate (panel d): the log-variance of the compression term grows, which down-weights a loss whose floor is already handled by free bits, while the local term is up-weighted as the hardest of the four to satisfy. Without centring none of this happens, and the same configuration collapses within the first ten percent of training.

C. Document homogeneity

The centring requirement is a property of the corpus, not of the method in general. The probe reported in Section III, a median pairwise cosine of 0.9997 over 200 documents, quantifies how little the raw encoder distinguishes one judgment from another. On a corpus whose documents already occupy distinct regions of the embedding space, the teacher target carries enough variation on its own and the predictor asymmetry alone would likely suffice. Legal text is at the opposite extreme, and we expect any non-contrastive document objective to need an explicit centring or sharpening step on such data.

VI. Discussion

The result that organises the paper is the crossover in Figure 2. A compression objective and a reconstruction objective produce representations with different inductive biases, and the better choice depends on how much supervision the downstream task will receive. Under ten examples per article the compressed representation is more linearly separable and far more stable, which is exactly what a low-data practitioner needs. Past fifty examples the reconstruction representation provides richer features that a linear head can exploit, and the compression that helped earlier now removes useful signal. Reading the bottleneck as a learned prior makes both halves of this behaviour expected rather than contradictory.

The stability finding deserves its own emphasis. Mean macro-F1 at $N = 10$ is low for every method, as it must be with ten examples and ten imbalanced labels, but the spread differs sharply. A method that returns zero on forty percent of seeds is not usable in a setting where one cannot afford many annotation rounds, and on this axis

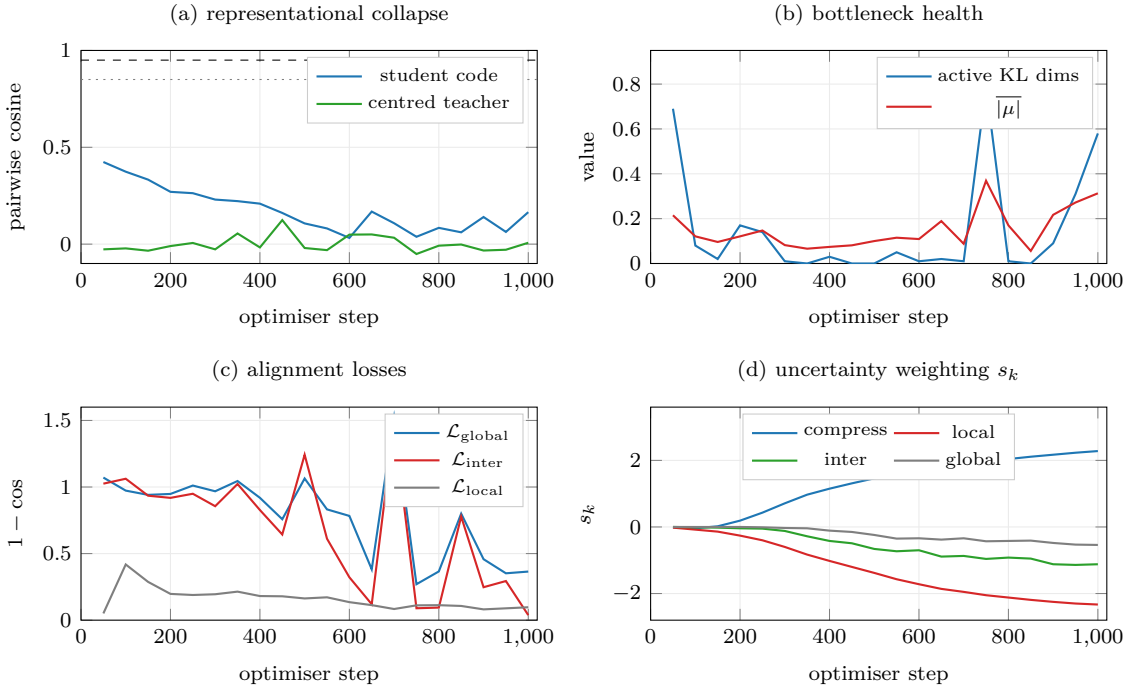


Fig. 3. First epoch of GlocalDoc pre-training after the centring fix. (a) The collapse metric stays far below the alarm (0.85, dotted) and stop (0.95, dashed) thresholds, and the centred teacher target has near-zero pairwise cosine. (b) The bottleneck remains informative as β warms up, with the fraction of active latent dimensions and the mean magnitude of μ both rising. (c) The document-level alignment losses decrease despite the growing compression weight. (d) The uncertainty weighting log-variances separate, which shows the weighting is active rather than stuck at its uniform initialisation.

GlocalDoc is clearly preferable even where the mean gap to H-MLM is small.

Several limitations bound these conclusions. We report one pre-training run per objective, so the variance we measure is over fine-tuning seeds and not over pre-training randomness; a pre-training seed could move the crossover point. The backbone is a distilled six-layer encoder and the pre-training is short, which keeps the experiments affordable but leaves open whether a larger model or a longer schedule would let the bottleneck retain more of the detail it currently discards. The stabilisation mechanisms were introduced as a sequence of diagnostic interventions against observed collapse rather than as a controlled ablation, so we can state that the combination trains and that centring is the decisive component on this corpus, but we cannot yet attribute precise contributions to the variance, covariance, and temporal terms. All results are on a single dataset in a single legal jurisdiction.

VII. Conclusion

We introduced GlocalDoc, a pre-training framework that compresses a corrupted, partially observed document through an information bottleneck while aligning it to a momentum teacher at the paragraph, partial-document, and full-document level. On few-shot classification of ECtHR cases the framework is preferable to a hierarchical masked language model only in the smallest-data regime, where it is both more accurate and far more stable across

seeds, and it is outperformed once fifty or more labelled cases per article are available. We traced a representational collapse to the near-constant teacher target produced by boilerplate-heavy legal text, and showed that centring the target removes it. Future work should run controlled ablations of the stabilisation terms, test a decoupled or annealed compression schedule that preserves more lexical detail at larger N , and study whether the crossover point can be shifted with a larger backbone or a curriculum over the number of labelled examples.

Appendix A Loss details

The compression term is the closed-form Kullback-Leibler divergence between the diagonal Gaussian posterior $\mathcal{N}(\mu, \text{diag}(\sigma^2))$ and the standard normal prior, computed from $\log \sigma$ for numerical stability under bfloat16. The per-dimension free-bits floor replaces each term by its maximum with λ , so a dimension contributes no gradient while it uses fewer than λ nats. With $\lambda = 0.05$ and a 256-dimensional latent the floor on the total divergence is 12.8 nats, low enough to permit substantial compression while ruling out a full posterior collapse to the prior. The cosine objectives use $d(u, v) = 1 - \cos(u, v)$; $\mathcal{L}_{\text{local}}$ flattens the per-document paragraph matrices before computing the mean cosine, so each student paragraph is matched to the teacher paragraph at the same position. The uncertainty weighting follows the homoscedastic form of [23], where

TABLE II
Configuration. Top: shared. Middle: GlocalDoc. Bottom: H-MLM and fine-tuning.

Setting	Value
Backbone	DistilRoBERTa-base (6 layers, 768)No pre.
Max paragraphs / chunk size	50 / 510 tokens
Precision / optimiser	bfloat16 / AdamW
Gradient clip	1.0
GlocalDoc epochs / eff. batch	5 / 8
Encoder LR / weights LR	3×10^{-5} / 1×10^{-2}
Schedule / warm-up	linear / 10%
EMA rate τ	0.996
β final / warm-up fraction	1.0 / 0.25
Free bits λ / latent dim	0.05 nats / 256
Projector / predictor	256-512-768 / 768-512-768
Variance / covariance weight	5.0 / 0.04
Variance hinge γ	0.5
Temporal buffer / var / cov	16 / 25.0 / 1.0
Centre momentum	0.9
Sentence / span / paragraph mask	0.20 / 0.15 / 0.30
H-MLM epochs / eff. batch	3 / 4
H-MLM LR / loss weight	5×10^{-5} / 0.5
H-MLM mask rate	0.15
Fine-tune epochs / LR	10 / 2×10^{-5}
Fine-tune schedule / threshold	cosine, 10% / 0.5
N / seeds	{10, 50, 100} / {0,...,4}

each learned s_k is the log-variance of a Gaussian likelihood and the term s_k acts as a regulariser that prevents the weights from growing without bound.

Appendix B Diagnosing collapse

The stabilisation mechanisms were added one at a time in response to distinct failure modes. A shared stop-gradient encoder collapsed because it could satisfy every alignment loss by becoming input-invariant; replacing it with a separate momentum teacher encoder and a student-side predictor removed that fixed point and let the encoder move. The bottleneck then collapsed downstream, with diverse paragraph vectors mapped to a constant document code; temporal variance and covariance terms on the document code and on the bottleneck mean addressed this. Aggressive free bits at $\lambda = 0.5$ over-constrained the mean and were reduced to $\lambda = 0.05$. The decisive failure was the near-constant teacher target induced by boilerplate, measured as a median inter-document cosine of 0.9997, which DINO centring removed. The first stable epoch is the one shown in Figure 3.

Appendix C Hyperparameters

Table II lists the configuration of both pre-training objectives and the shared fine-tuning procedure.

Appendix D Per-seed results

Table III gives the macro-F1 of every run. The two zero entries for H-MLM at $N = 10$ are the runs that fail to

TABLE III
Per-seed macro-F1 on the test split.

Method	N	s0	s1	s2	s3	s4
H-MLM	10	0.000	0.000	0.000	0.001	0.003
	50	0.017	0.033	0.031	0.027	0.016
	100	0.037	0.027	0.037	0.063	0.047
	10	0.012	0.000	0.001	0.028	0.054
	50	0.265	0.287	0.291	0.178	0.344
	100	0.390	0.376	0.395	0.317	0.383
GlocalDoc	10	0.042	0.035	0.040	0.049	0.057
	50	0.093	0.094	0.151	0.103	0.101
	100	0.184	0.191	0.178	0.188	0.177

learn, and they account for most of the high standard deviation reported for that setting in Table I.

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